DOKUZ EYLUL UNIVERSTY

ENGINEERING FACULTY

DEPARTMENT OF COMPUTER ENGINEERING

CME 2210 OBJECT ORIENTED ANALYSIS AND DESIGN

PHASE 1

FACTORY MANAGEMENT SYSTEM

ANALYSIS AND REQUIREMENTS DEFINITON REPORT

by

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System Definition and Objectives

Factory management system is a software application which integrates manual labor duties, management responsibilities and optimizes them via automation.

The system will basically improve real-time observation, production tracking, resource distribution and workforce management while aiming to improve efficiency, reduce manual workload, boost overall productivity throughout the factory and assist decision-making.

This system will be built according to OOP principles which will provide modularity, scalability and maintainability.

Functional Requirements

Functionalities are described based on user types.

Logging in as employer or manager;

A manager in the Factory Management System is responsible for overseeing factory operations, ensuring efficiency, and making strategic decisions. The system provides managers with various tools to monitor production, manage employees, and optimize resources.

- Employee rating and performance review
- Production management
- Shift management
- Stock management
- Customer and order management
- Financial monitoring

Logging in as employee:

An employee in the Factory Management System uses the system to track their work-related information, manage schedules, and stay updated on payroll, performance, and leave balances. The system provides employees with tools to enhance productivity and maintain transparency in their work environment.

- Payroll monitoring
- Review insurance information
- Managing leave rights
- Display work rating
- Display shift information

Logging as a customer:

A customer in the Factory Management System utilizes the platform to place and manage orders, track their status, and access product information. The system makes the customer experience smooth and transparent by offering essential tools for managing their purchases and providing feedback.

- Placing and managing orders
- Reviewing order status
- Feedback system
- Reviewing stock information about certain products

Non-Functional Requirements

1. Performance

Response Time: The system should respond to user requests within 2-3 seconds under normal conditions.

Throughput: The system should be able to handle a minimum of 500 users at the same time without downgrade on performance.

Load Handling: The system must be able to scale and handle up to 10,000 records in real-time for inventory, orders, and employee data.

2. Scalability

Horizontal Scaling: The system should be able to scale horizontally to avoid facing problems in case of increasing number of users and data.

Database Scalability: The database should support both vertical and horizontal scaling to manage growing amounts of operational data.

3. Security

Role-Based Access Control: The system must ensure that users can only access data relevant to their roles.

4. Maintainability

Modularity: The system should be modular and organized update easily, bug fixes, and new feature integrations without breaking the whole system.

Code Quality: The system should follow high quality codes and be properly documented to apply easy maintenance.

Use-Cases

Use Case 1: Log In to View Payroll and Shift Information

Participating Actor: Employee

Entry conditions:

- The employee has an active account in the system.
- The employee knows their login credentials (username and password).

Main Flow:

1. The employee navigates to the system's login page.
2. The employee enters their username and password.
3. The system authenticates the employee.
4. The system displays a menu with options to view payroll, insurance information, and shift schedule. The employee returns to menu.
5. The employee chooses Payroll, where they can view the salary details and payment history. The employee returns to menu.
6. The employee chooses Shift Information, where they can view their assigned shifts for the next week. The employee returns to menu.
7. The employee chooses Insurance Information to review it’s information.
8. The employee logs out after reviewing the information.

Exit conditions:

- The employee has viewed their payroll, shift details, and insurance information.

Use Case 2: Review Employee Performance and Approve Leave Request

Participating Actor: Manager

Entry conditions:

- The manager has logged in with the necessary role-based access.
- The employee has submitted a leave request through the system.

Main Flow:

1. The manager logs into the system using their password and username.
2. The manager moves to the Employee Performance section and selects an employee.
3. The system displays the employee's performance report.
4. The manager reviews the employee's work ratings, feedback, and performance datas. The manager moves back to the menu.
5. The manager navigates to the Leave Requests section and reviews the pending leave request from the employee.
6. The manager approves or rejects the leave request based on the employee's performance and work requirements.
7. The system updates the employee's leave status and sends a notification to the employee.

Exit conditions:

- The employee's leave request is processed and updated in the system.
- The employee's performance review is recorded and can be seen by the employee and manager.

Use Case 3: Place and Track an Order

Participating Actor: Customer

Entry conditions:

- The customer has an active account.
- The customer is logged in.

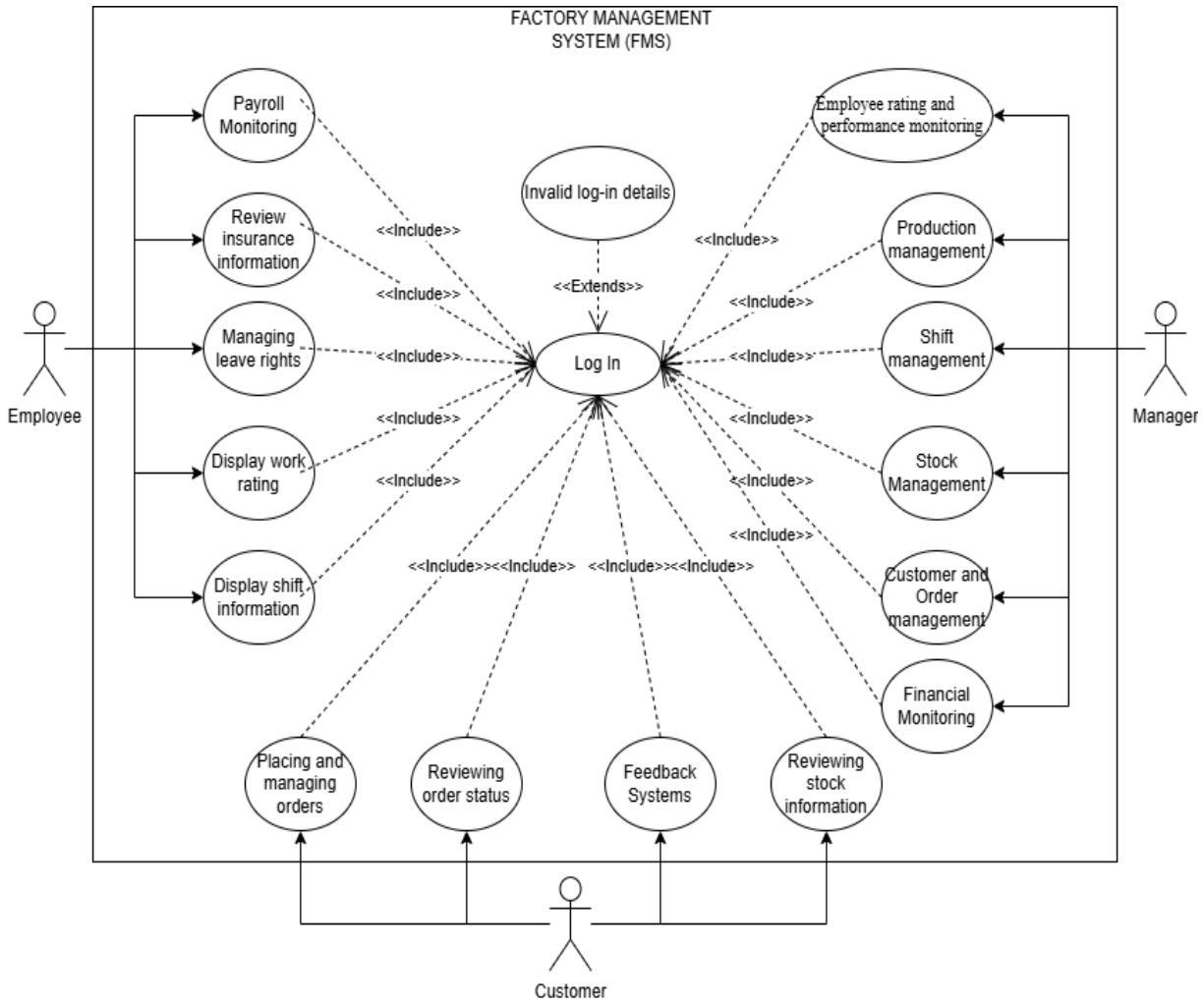
Main Flow:

1. The customer logs into the system.
2. The customer searches for available products and selects an item to purchase.
3. The customer proceeds to the checkout page, providing necessary details.
4. The system confirms the order and generates an order ID.
5. The customer navigates to Order Status to track the current status of their order.

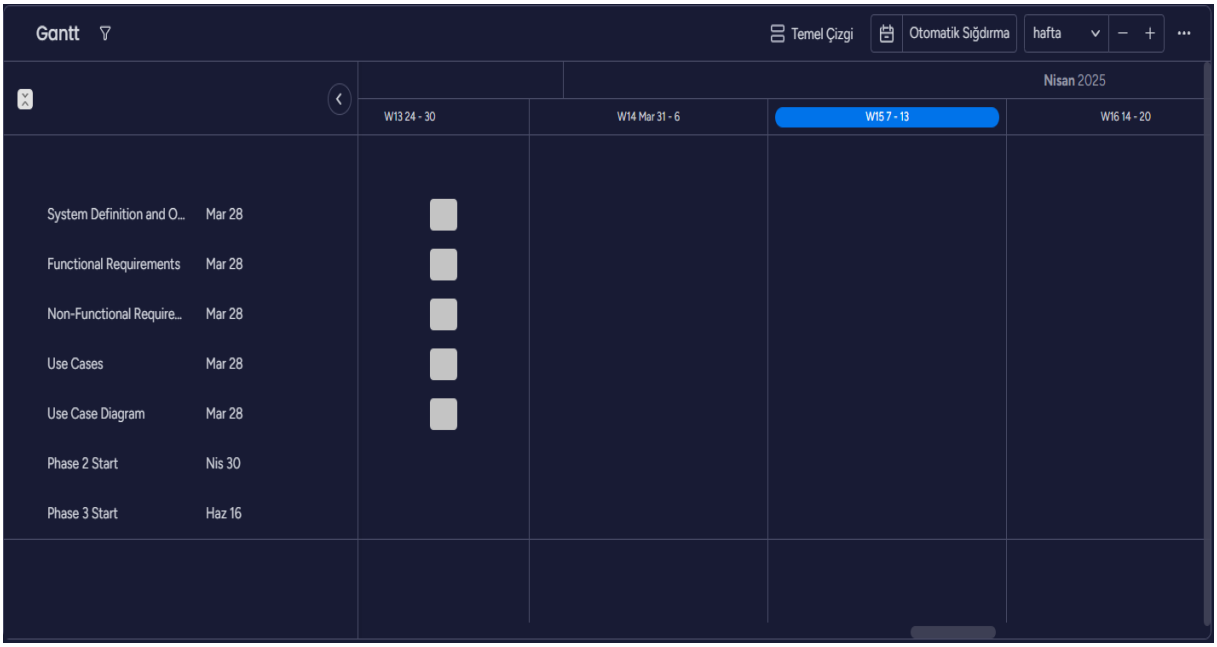
Exit conditions:

- The customer's order is placed, and they can track its status in real-time.

Use Case Diagram



Gantt Chart


Gantt
Temel Çizgi
Otomatik Sığdırma
hafta
Nisan 2025
W13 24 - 30
W14 Mar 31 - 6
W15 7 - 13
W16 14 - 20
System Definition and O...
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Functional Requirements
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Non-Functional Require...
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Use Cases
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